

Guidelines

- Kindly use the given template for submitting your prototype(Make a copy of the template)
- One team is only required to submit one project.
- Follow this template while preparing your presentation.
- You are welcome to add as many POCs and design concepts to support your project.
- The project should be feasible and the team members should be capable enough, to come up with the fully functioning prototype of the same idea, if required.

The logo for unilog | IIS, featuring the word 'unilog' in blue and 'IIS' in white, separated by a vertical line. The background of the slide is a dark, futuristic server room with glowing blue and orange light trails.

UniHack

AI-Powered Product Intelligence for
Industrial Commerce

Team Details

Team Name: Tolerance

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Executive Brief: Tolerance

- Core Problem: Industrial distributor catalog data is highly cryptic (e.g. '3/8 CPLG BRS 150#'), misspelled, and lacks structure, rendering products unsearchable and driving up catalog maintenance costs.
- Our Solution: An AI-powered product data intelligence pipeline. Combines deterministic parsing rules for speed and accuracy with LLM-assisted classification, structured attribute extraction, and validation.
- Key Innovation: Exposes confidence scores and provenance tags (Rule-Based vs LLM-Inferred) for every single specification, providing a transparent audit trail for human catalogers.

Technical Challenge Answers

1. Understanding Limited Info (Enrichment):

- We utilize a hybrid model. Clean dimensions and materials are tokenized using fast deterministic regex patterns. Ambiguous text fragments are parsed using Gemini Flash by supplying the category context.

2. Accuracy & Trust (Validation):

- Tolerance runs sanity validation rules (e.g., flagging PVC material with class 150 metal ratings). Missing fields are filled using Gemini's general domain knowledge, labeled clearly as 'AI-Inferred' for easy auditing.

3. Scalability (Enterprise Catalog Engine):

- Built as a concurrent chunked batch pipeline. Repeat manufacturer name lookups are cached locally. In production, this scales via an asynchronous job queue (BullMQ/Redis) with multiple background workers.

Opportunities & USP

- Frictionless Audit Trail: Traditional AI classifiers behave as black boxes. Tolerance displays confidence margins and source provenance tags directly in the cataloger UI, enabling rapid auditing.
- Active Conflict Checking: Rather than outputting corrupt values, Tolerance flags logical contradictions (such as incompatible fitting sizes and thread types).
- Rate-Limit Immunity: Pre-cached seed catalog routes guarantee 100% demo availability and rate-limit immunity during live evaluation.

Features Offered by the Solution

- Fuzzy Manufacturer & Brand De-duplication: Reconciles raw spelling errors (e.g. 'lasco fittings ltd').
- 3-Level Deep Parts Taxonomy Classification: Places items into a structured category path using Gemini Flash.
- Structured Attribute Extractor: Extracts key-value-unit specifications.
- UOM & Fraction Standardization: Spacing, casing, and decimal-to-fraction conversions.
- E-commerce Description Builder: Generates CAPS invoice descriptions (≤ 40 chars) and mobile descriptions (60-80 chars).
- ROI Evaluation Module: Tracks data density gains and calculates manual analyst hours saved.

Process Flow Diagram

[Raw Messy Product Row] --> [Schema Profiler]

--> [Manufacturer & Brand De-duplication (Fuzzy Mapper)]

--> [Taxonomy Classifier (Gemini Flash)]

--> [Hybrid Attribute Extraction (Regex + LLM)]

--> [Validation & AI Enrichment (Conflict Check & Defaults)]

--> [UOM Normalization & Cleansing (Standard Spacing)]

--> [Invoice & Mobile Description Builder]

--> [Output: Enriched JSON/CSV + Business ROI Dashboard]

Wireframes/Mock diagrams of the proposed solution (optional)

Proposed Solution Architecture

- Client Dashboard: Next.js 15, TypeScript, Tailwind CSS v4, Lucide Icons.
- API Gateway: Next.js API Routes (/api/enrich) handling JSON payload processing.
- Core Engine: Structured TypeScript utility pipe coordinates all modules sequentially.
- LLM Provider: Google Gemini API via @google/generative-ai routing to gemini-flash-latest.
- Infrastructure: Vercel (Hobby Tier) for live public hosting.

Technologies Used

- Next.js 15 & React 19 (Serverless Web Framework)
- TypeScript (Type-safe business logic)
- Tailwind CSS v4 & PostCSS (Modern technical styling)
- Google Gemini Flash SDK (AI-assisted taxonomy & extraction)
- Vercel Platform (Continuous deployment & hosting)

Estimated Implementation Cost

- LLM Cost: ~₹0.006 per SKU (based on Gemini API pricing at ₹83 per USD)
- Database & Storage: ₹0.00 (leveraging lightweight local caching)
- Cloud Hosting: ₹0.00 (Vercel Hobby Tier serverless deployment)
- Total cost for MVP testing: ₹0.00

Snapshots of the MVP

- Engineering Blueprint Transformation View: Floating schematic cards and leader lines.
- Specification Audit Trail: Interactive table mapping confidence and provenance.
- Business ROI Metrics Panel: Analytics cards showing manual hours saved.
- Wireframe Schema Diagram: Saved in `public/dashboard_wireframe.jpg` for review.

Future Development & Roadmap

- ERP Connectors: Directly stream enriched data into popular distributor ERPs (e.g. Epicor P21, Infor, SAP).
- Digital Asset Sourcing: Automated scraping agents to locate and download PDF manual sheets and catalog drawings.
- Collaborative Human-in-the-loop (HITL): UI dashboard for catalogers to approve, reject, or correct AI extractions, retraining model rules in real-time.

Final Submission Links

GitHub Public Repository:

- <https://github.com/Saatvik-G/Tolerance.git>

Working Prototype URL:

- <https://tolerance-ten.vercel.app/>

Demo Video Link (3 Minutes):

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<https://app.guidde.com/share/playbooks/8iiUdixCsLkudTRbbf8Jws?origin=PBBuMYerseb1XleJip2pj6eLYmS2&mode=videoOnly>

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Thank You

